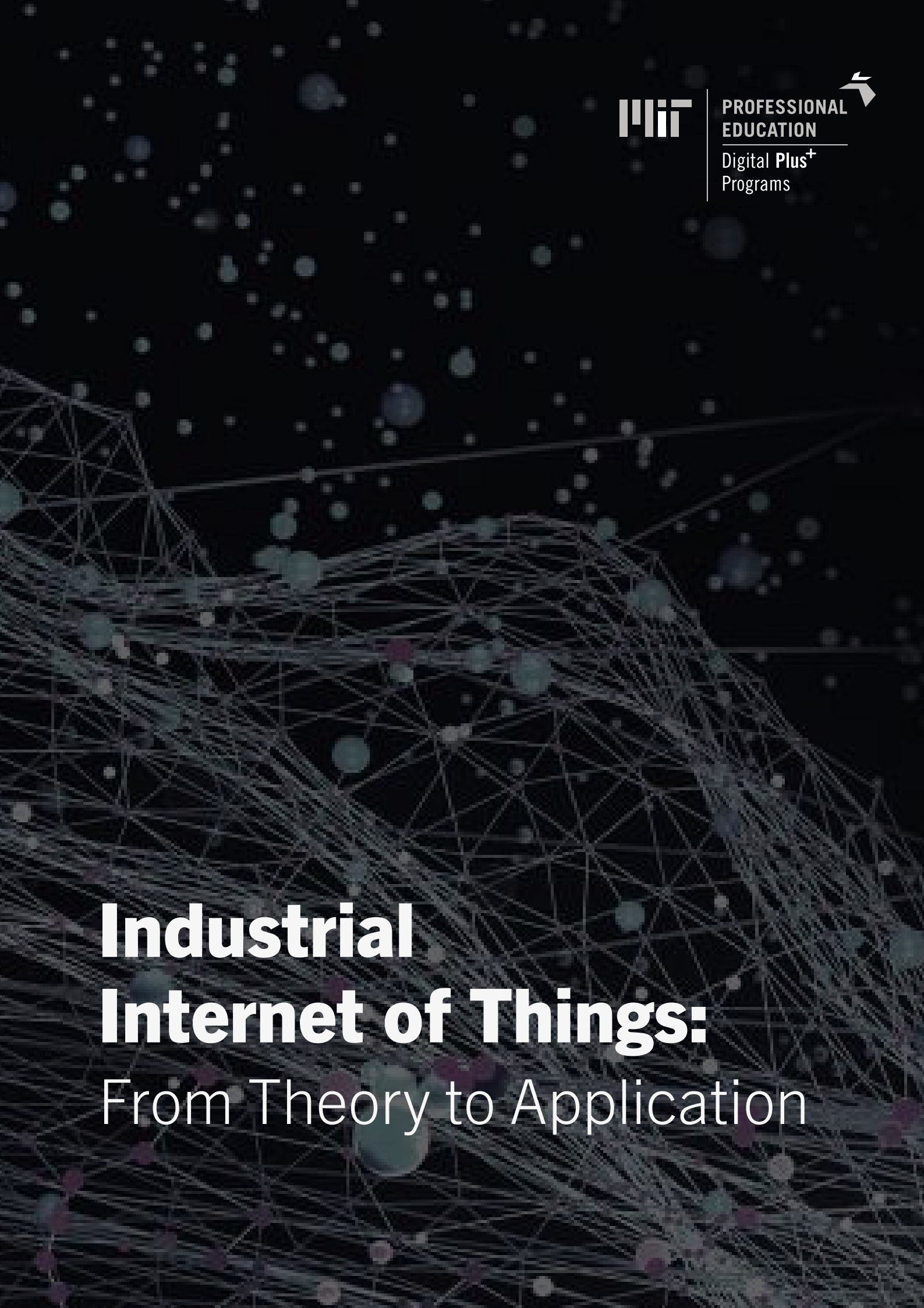
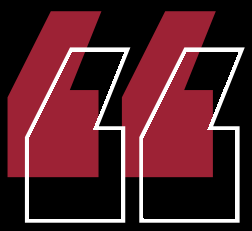


PROFESSIONAL EDUCATION
Digital Plus+ Programs

Industrial Internet of Things:
From Theory to Application



At their heart, IoT systems are characterized by the convergence of micro-sensing, computation, and communication.

Brian W. Anthony

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Technology Today



The Internet of Things (IoT) is taking the way we live and work to a whole new level. This technology allows anything from sensory devices to household appliances be connected and interact through the internet, without any need for human intervention.

Welcome to the future of organizations.

These exhilarating conditions require leaders who adapt, learn continuously, and understand that **technology is the key to taking their organizations to a more beneficial**, optimized, and connected level.

Benefits of improving your digital strategy

US$ billion 3,5

The number of cellular IoT connections is expected to reach 3.5 billion by 2023.

Source: Ericsson

80%

On average, over 80% of senior executives from all industries say that the IoT is fundamental to either some or all lines of their business.

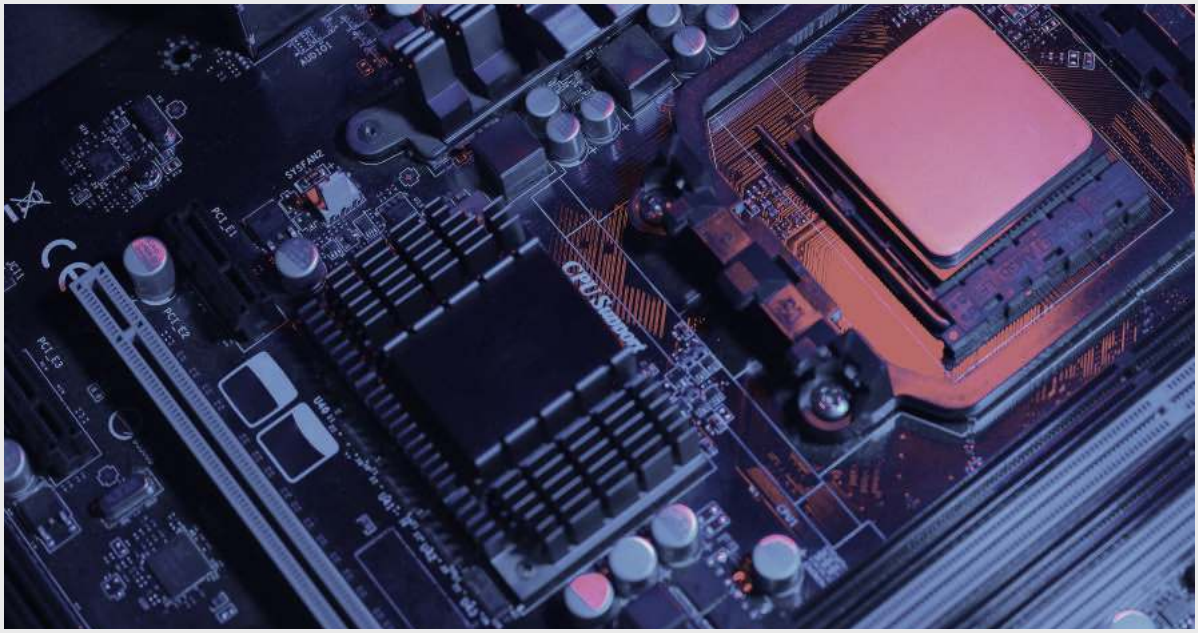
Source: Statistics

US$ billion 64

Over 64 billion IoT devices are expected worldwide by 2025.

Source: Business Insider, IoT Analytics, Gartner, Intel

About this Course



Throughout this course, you will study the main detection, computing, and communication software technologies that are central to the recent stream of IoT+ practices.

The technology that allows you to detect, reason and make automated decisions is deeper and more complex than the mere Internet of Things (IoT). In this course, we will be introducing you to the technological concept that goes beyond IoT: IoT+.

During the course, you will:

1. **Develop your skills while overcoming technological challenges.**
2. **You will gain insight into relations between the real and artificial world.**

This will allow you to play a positive role over people, machines, environments and eco-systems, which you will also implement in your professional field

9 weeks | 8-10 hours a week | Online

Key Takeaways

Throughout this course you will develop the following skills:

01 **Design and assembly of integrated systems,** sensors, and IoT+ in which hardware and software interact.

02 **Selection and definition** of teams or components that are available on the market.

03 **Designing IoT+ products** that will help you reach your hardware and software's full potential.

04 **Solutions to Data Science problems** using modern Machine Learning techniques.

05 **Industry solutions** (transport, health, agriculture, etc.) using IoT+

06 **Construction of complete hardware and software systems** while being exposed to new and exciting hardware prototyping platforms.

07 **Design and implementation** of telecommunication networks for the use of IoT+.

08 **Utilization of sensors,** detection technologies, and integrated systems in the IoT+ setting.

09 **Through IoT+**, you will use the technology at your disposal to develop and improve real applications with intelligent technologies.



Participant Profile

This course is for those who wish to become certified professionals in the IoT+ sector. Professionals who will benefit the most from the knowledge and skills acquired in this course are:

Engineers and engineering managers

Design and manufacturing engineers who are looking to learn about data and modeling in the manufacturing industry.

Data scientists who want to put their skills into practice in expanding fields like smart manufacturing.

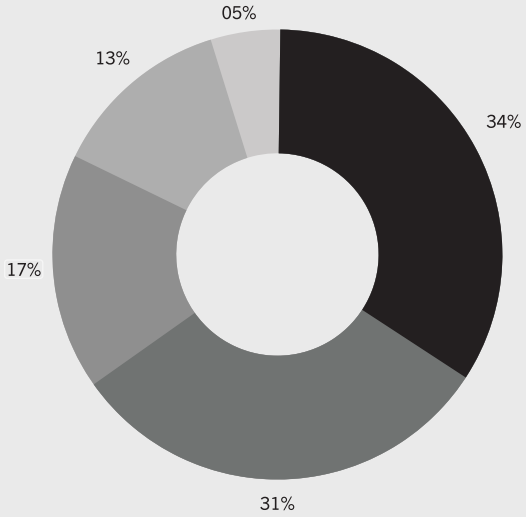
Consultants who aim to add value to the latest technological changes in the manufacturing process

To enroll in this course **it is not necessary to have previous knowledge of the material.** There are activities where knowing programming languages (MATLAB and/or Python) can help you, **but it is not necessary.**

Knowledge of physics and mathematics can help you perform the theoretical activities of measurement and data collection with sensors. But an advanced knowledge of physics and mathematics is **not required.**

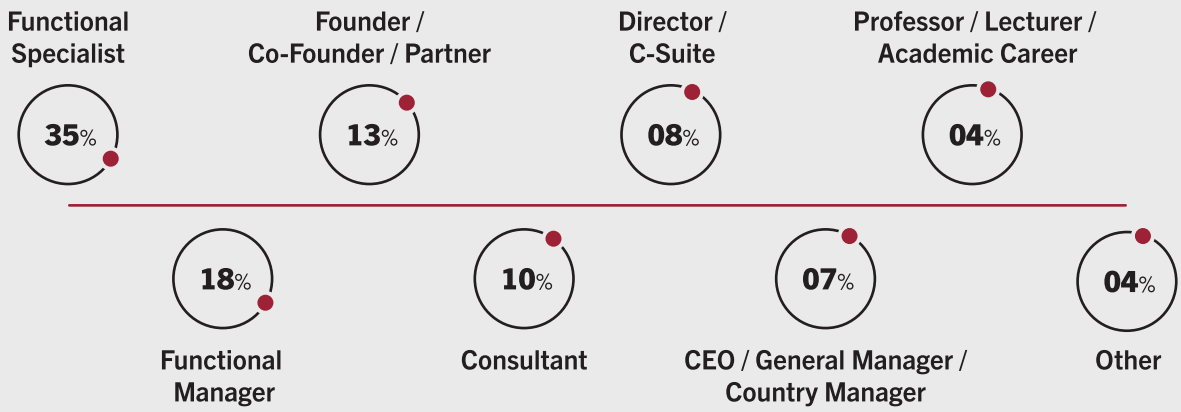
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Years of Experience



10-15 years	34%
5-9 years	31%
More than 25 years	17%
16-25 years	13%
Less than 5 years	05%

Job Category



Industry

Engineering	12%	Telecommunication	12%
Technology	12%	Consumer Goods & Services	12%
Manufacturing	10%	Agriculture	10%
Consulting	06%	Energy	06%
IT / Data	05%	Food & Beverage	05%
Education	05%	Construction	05%
Computer / Electronics	05%	Mining	05%
Automotive	04%	Other	04%

Instructors



Dr. Brian W. Anthony

Director of the MIT Master of Engineering in Advanced Manufacturing and Design. Associate Director, MIT.nano

Dr. Anthony is the director of the MIT Master of Engineering in Advanced Manufacturing & Design, and codirector of the Medical Electronic Device Realization Center. He has more than 20 years of experience in product realization and is an **Emmy award winner** in the Academy of Television Arts & Sciences for his innovation in broadcast technology.

Dr. Anthony designs instruments and techniques to monitor and control physical systems. He also uses mechanical, electrical, and optical engineering as well as computer science and optimization to develop solutions.
Dr. Anthony's research revolves around the design of instruments and techniques for measuring and controlling complex physical systems. It includes the development of instruments and measurement solutions for manufacturing, medical diagnostics, and imaging diagnostics. In addition to his academic work, he has **extensive experience in market-driven technological innovation, product development and entrepreneurship, and marketing** at the point where information technology and advanced manufacturing come into contact. As an educator, his main interests are the modeling of large-scale systems dealing with decision-making and the development of optimization algorithms and software to analyze and design these systems. He is an expert in market-driven technological innovation as well as in entrepreneurship.

Dr. Anthony holds the following degrees:
• PhD in Engineering, Massachusetts Institute of Technology
• MS in Engineering, Massachusetts Institute of Technology
• BS in Engineering, Carnegie Mellon University broadcast technical innovation.

"Using technology to refine the manufacturing process has been a steady interest throughout my career and allows better products to be offered on the market, faster and more economically."

Course Outline

Module 0
Introduction, Markets, Needs and Trends

1. What Is IoT and What Is There Beyond IoT?
2. Market Sectors
3. Needs and Use Cases
4. Platforms

Module 1
Sensors Part 1

1. Selected Needs: A Complete Case
2. First Data Exploration and Preliminary Display
3. Detection System Contact
4. Introduction to Sensors
5. Choosing a Sensor.

Module 2
Sensors Part 2

1. Data Acquisition
2. Fourier Transform
3. Time and Frequency
4. Basic Analysis and a Complex Case

Module 3
Computing

1. Algorithms
2. Computing and Learning
3. Convolution
4. Power to Operate
5. Statistics
6. Cases with Statistics

Module 4
Learning

1. Automated Learning Methods
2. Decision Trees
3. Joint Learning Part I
4. Joint Learning Methods Part II
5. Support Vector Machines
6. Neural Networks

Module 5
Localization

1. Introduction to Localization
2. Waves
3. GPS
4. Inertial Detection
5. SLAM Technology

Module 6
Communication

1. Introduction to Communication Detection Technologies

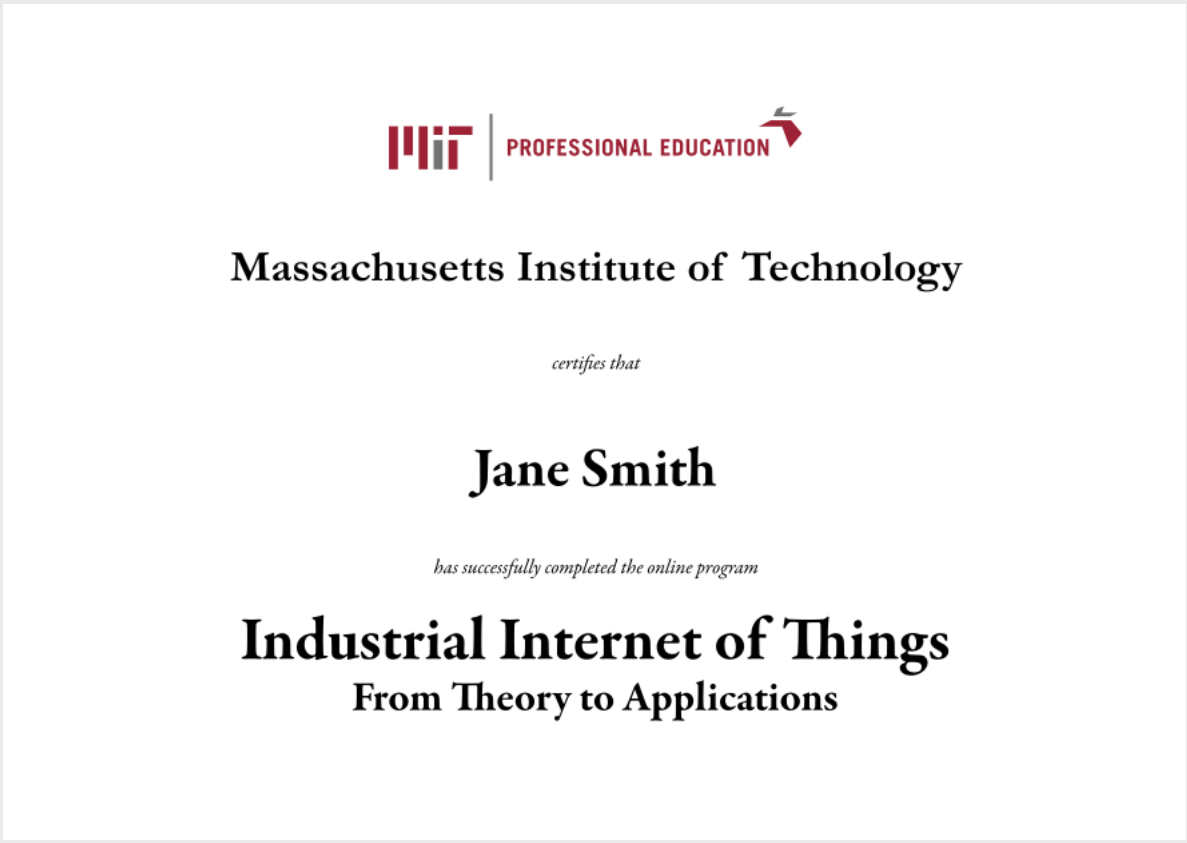
Module 7
Integration Part I and II

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Certificate of Completion

All participants who successfully complete the course will receive an MIT Professional Education Certificate of Completion and 7.2 Continuing Education Units (CEUs*).

In order to obtain CEUs, participants must complete a required CEU accreditation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

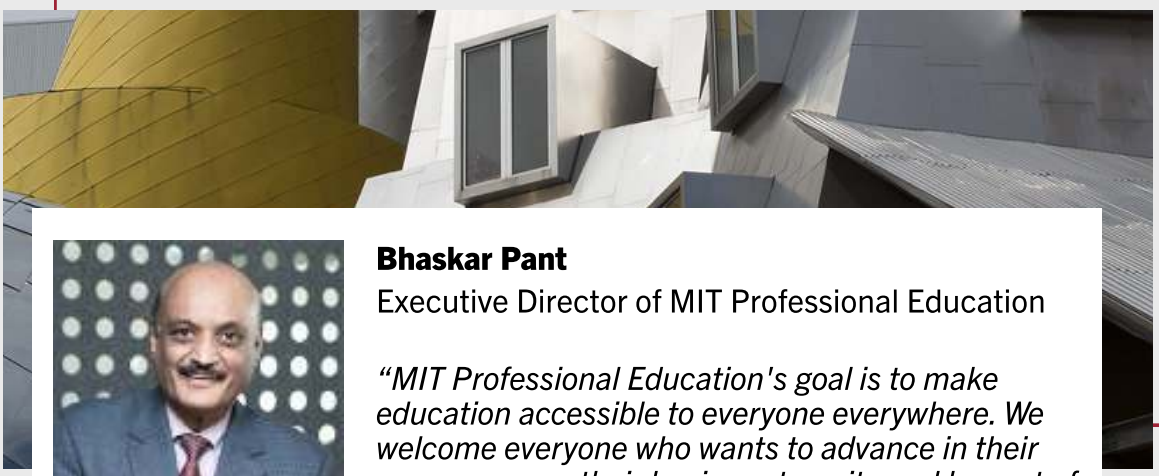
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

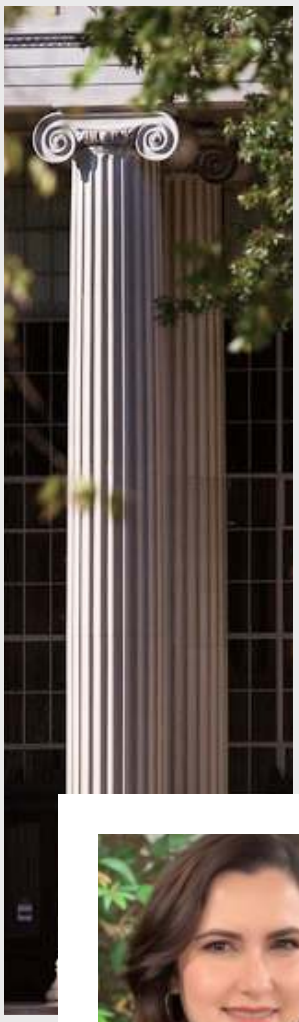


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience.**

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Director of Global Programs at MIT Professional Education

"The Internet of Things is here to stay. As we embark upon Digital Transformation and Industry 4.0, we are witnessing increasing applications for the Internet of Things in professional settings to maintain organizations' competitive edge. IoT helps people live and work smarter, as well as gain increasingly more control over their lives."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **15% tuition reduction** for MIT Professional Education online courses and short-duration, in-person programs.
2. **Continued access to the virtual campus for six months after completion** of the course.
3. **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network.**
4. Invitation to the **MIT Professional Education Alumni LinkedIn group.**
5. **Networking opportunities** with other MIT Professional Education Alumni.

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PROFESSIONAL EDUCATION
Digital Plus+ Programs

Are you ready to take the lead on the latest technologies?

Frequently Asked Questions

global alumni
-excellence or nothing-

For more information, please contact us:

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Email: **professionalprograms@mit.edu**

Website: **https://professionalprograms.mit.edu**